

## Qualification Pack



# IoT Installation Solution Architect

QP Code: TEL/Q6216

Version: 3.0

NSQF Level: 5

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## Qualification Pack

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## Qualification Pack

### TEL/Q6216: IoT Installation Solution Architect

#### Brief Job Description

The individual in this job is responsible for conducting the site survey for layout, planning and designing for installation and deployment of IoT sensors/devices, IoT gateways and access layer connectivity in IoT ecosystem to suggest and implement the best IoT solution for the business model.

#### Personal Attributes

The individual needs to have the ability to upgrade skills with changing technologies, work in a team, multitask and track multiple projects simultaneously with full dedication and willingness. The individual should have generic communication and leadership skills, attention to details, excellent problem-solving capabilities, strong quantitative abilities and good interpersonal skills. The individual with skills like innovation, customer focus, collaboration, value creation and professionalism would have added advantage in this job role.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [TEL/N6260: Perform Market Analysis on Application of IoT](#)
2. [TEL/N6261: Supervise in Installation IoT Devices and System](#)
3. [TEL/N6262: Administer Acceptance Testing and Site Optimization Activities](#)
4. [TEL/N6280: Maintain Secure IoT Network Infrastructure](#)
5. [TEL/N9103: Implement Effective Interaction at workplace](#)
6. [TEL/N9104: Manage Work, Resources and Safety at workplace](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

#### Qualification Pack (QP) Parameters

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Sector</b>     | Telecom                           |
| <b>Sub-Sector</b> | Network Managed Services          |
| <b>Occupation</b> | Network Operation and Maintenance |
| <b>Country</b>    | India                             |

## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NSQF Level</b>                                         | 5                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Credits</b>                                            | 18                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/3114.6216                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Minimum Educational Qualification &amp; Experience</b> | Completed 2nd year of UG (UG Diploma) (of 3-year/ 4-years UG)<br>OR<br>Completed 2nd year diploma after 12th<br>OR<br>12th grade Pass with 2 Years of experience Relevant Experience<br>OR<br>Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience Relevant Experience<br>OR<br>Previous relevant Qualification of NSQF Level (4) with 3 Years of experience Relevant Experience |
| <b>Minimum Level of Education for Training in School</b>  | 12th Class                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Minimum Job Entry Age</b>                              | 18 Years                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Next Review Date</b>                                   | 30/06/2028                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>NSQC Approval Date</b>                                 | 19/08/2025                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Version</b>                                            | 3.0                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Reference code on NQR</b>                              | QG-05-TL-00470-2025-V2-TSSC                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>NQR Version</b>                                        | 2                                                                                                                                                                                                                                                                                                                                                                                                            |

## Qualification Pack

### TEL/N6260: Perform Market Analysis on Application of IoT

#### Description

This OS unit analyzes and evaluates market opportunities for IoT applications, gathering industry trends, segmenting markets, assessing competitors, understanding customer needs, and creating insights for strategic decision-making.

#### Scope

The scope covers the following :

- Market research and data collection
- Competitor and trend analysis
- Customer need mapping and value proposition identification
- Reporting, recommendations, collaboration, and compliance

#### Elements and Performance Criteria

##### *Market research and data collection*

To be competent, the user/individual on the job must be able to:

- PC1.** identify relevant IoT domains (e.g., smart home, industrial IoT, health tech) and collect secondary data from industry reports, white papers, publications, and digital sources.
- PC2.** conduct primary research by designing surveys, interviews, or questionnaires to capture end-user needs, market gaps, and technology adoption barriers.
- PC3.** extract and organize relevant data using appropriate tools (Excel, Google Sheets, databases) for effective analysis and interpretation.
- PC4.** apply classification techniques to segment IoT markets based on factors like industry type, geography, application, and customer size.

##### *Competitor and trend analysis*

To be competent, the user/individual on the job must be able to:

- PC5.** benchmark existing IoT solutions and vendors in terms of features, pricing, performance, scalability, and support.
- PC6.** identify current trends in IoT technologies, such as edge computing, AI integration, 5G connectivity, and regulatory developments.
- PC7.** analyze SWOT (Strengths, Weaknesses, Opportunities, Threats) of top competitors to assess positioning and market entry points.
- PC8.** prepare visual representations (charts, heat maps, graphs) of market shares, growth rates, and emerging trends using tools like PowerPoint, Canva, or Tableau.

##### *Customer need mapping and value proposition identification*

To be competent, the user/individual on the job must be able to:

- PC9.** interpret research findings to identify latent needs and unserved demands in the target market.
- PC10.** map customer pain points with potential IoT applications to determine areas of maximum value creation.

## Qualification Pack

- PC11.** recommend features, use cases, or deployment strategies that align with customer needs and business feasibility.

### *Reporting, recommendations, collaboration, and compliance*

To be competent, the user/individual on the job must be able to:

- PC12.** summarize market insights in a structured format, highlighting key findings, market gaps, and opportunity areas.
- PC13.** draft detailed market analysis reports with strategic recommendations for product planning, investment decisions, and potential partnerships.
- PC14.** present findings and insights effectively to stakeholders using storytelling techniques and data-backed arguments for clarity and impact.
- PC15.** coordinate with internal teams such as product development, marketing, and strategy to ensure market insights are aligned with business goals.
- PC16.** adhere to ethical standards, data privacy norms, and organizational protocols throughout the process of conducting and presenting market analysis.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies on market research, data sharing, and confidentiality.
- KU2.** roles and responsibilities of product, marketing, and strategy teams.
- KU3.** documentation procedures for research findings, reports, and presentations.
- KU4.** compliance norms regarding customer data privacy and third-party data usage.
- KU5.** organizational vision and how IoT solutions align with business objectives.
- KU6.** fundamentals of IoT architecture and applications across industries and technologies to enable them.
- KU7.** current technologies enabling IoT (e.g., sensors, cloud, AI, edge computing).
- KU8.** sources and techniques of primary and secondary market research.
- KU9.** methods of data segmentation and customer profiling.
- KU10.** approaches for competitor benchmarking and SWOT analysis.
- KU11.** tools and platforms used for data analysis, visualization, and reporting.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret industry reports, market data, and IoT-related documentation.
- GS2.** communicate findings and insights effectively with team members, clients, and stakeholders.
- GS3.** stay updated on current trends, innovations, and emerging technologies in the IoT domain.
- GS4.** manage research timelines and project schedules to deliver timely market analysis.
- GS5.** identify market opportunities and challenges through logical analysis and interpretation.
- GS6.** apply critical thinking to evaluate potential use cases and ROI of IoT applications across industries.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Market research and data collection</i>                                                                                                                                                   | <b>20</b>    | <b>12</b>       | -             | <b>2</b>   |
| <b>PC1.</b> identify relevant IoT domains (e.g., smart home, industrial IoT, health tech) and collect secondary data from industry reports, white papers, publications, and digital sources. | 5            | 3               | -             | 1          |
| <b>PC2.</b> conduct primary research by designing surveys, interviews, or questionnaires to capture end-user needs, market gaps, and technology adoption barriers.                           | 5            | 3               | -             | -          |
| <b>PC3.</b> extract and organize relevant data using appropriate tools (Excel, Google Sheets, databases) for effective analysis and interpretation.                                          | 5            | 2               | -             | 1          |
| <b>PC4.</b> apply classification techniques to segment IoT markets based on factors like industry type, geography, application, and customer size.                                           | 5            | 4               | -             | -          |
| <i>Competitor and trend analysis</i>                                                                                                                                                         | <b>20</b>    | <b>10</b>       | -             | <b>2</b>   |
| <b>PC5.</b> benchmark existing IoT solutions and vendors in terms of features, pricing, performance, scalability, and support.                                                               | 5            | 2               | -             | -          |
| <b>PC6.</b> identify current trends in IoT technologies, such as edge computing, AI integration, 5G connectivity, and regulatory developments.                                               | 5            | 2               | -             | -          |
| <b>PC7.</b> analyze SWOT (Strengths, Weaknesses, Opportunities, Threats) of top competitors to assess positioning and market entry points.                                                   | 5            | 2               | -             | 1          |
| <b>PC8.</b> prepare visual representations (charts, heat maps, graphs) of market shares, growth rates, and emerging trends using tools like PowerPoint, Canva, or Tableau.                   | 5            | 4               | -             | 1          |
| <i>Customer need mapping and value proposition identification</i>                                                                                                                            | <b>5</b>     | <b>10</b>       | -             | <b>2</b>   |
| <b>PC9.</b> interpret research findings to identify latent needs and unserved demands in the target market.                                                                                  | 2            | 5               | -             | 1          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> map customer pain points with potential IoT applications to determine areas of maximum value creation.                                             | 2            | 3               | -             | -          |
| <b>PC11.</b> recommend features, use cases, or deployment strategies that align with customer needs and business feasibility.                                   | 1            | 2               | -             | 1          |
| <i>Reporting, recommendations, collaboration, and compliance</i>                                                                                                | <b>5</b>     | <b>8</b>        | -             | <b>4</b>   |
| <b>PC12.</b> summarize market insights in a structured format, highlighting key findings, market gaps, and opportunity areas.                                   | 1            | 2               | -             | 1          |
| <b>PC13.</b> draft detailed market analysis reports with strategic recommendations for product planning, investment decisions, and potential partnerships.      | 1            | 2               | -             | 1          |
| <b>PC14.</b> present findings and insights effectively to stakeholders using storytelling techniques and data-backed arguments for clarity and impact.          | 1            | 1               | -             | 1          |
| <b>PC15.</b> coordinate with internal teams such as product development, marketing, and strategy to ensure market insights are aligned with business goals.     | 1            | 1               | -             | 1          |
| <b>PC16.</b> adhere to ethical standards, data privacy norms, and organizational protocols throughout the process of conducting and presenting market analysis. | 1            | 2               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                | <b>50</b>    | <b>40</b>       | -             | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                               |
|----------------------------|-----------------------------------------------|
| <b>NOS Code</b>            | TEL/N6260                                     |
| <b>NOS Name</b>            | Perform Market Analysis on Application of IoT |
| <b>Sector</b>              | Telecom                                       |
| <b>Sub-Sector</b>          | Network Managed Services                      |
| <b>Occupation</b>          | Network Operation and Maintenance             |
| <b>NSQF Level</b>          | 5                                             |
| <b>Credits</b>             | 4                                             |
| <b>Version</b>             | 3.0                                           |
| <b>Last Reviewed Date</b>  | 19/08/2025                                    |
| <b>Next Review Date</b>    | 30/06/2028                                    |
| <b>NSQC Clearance Date</b> | 19/08/2025                                    |

## Qualification Pack

### TEL/N6261: Supervise in Installation IoT Devices and System

#### Description

The OS oversees IoT device installation, ensuring compliance with technical specifications, infrastructure readiness, safety standards, and integration requirements, coordinating with field engineers, validating configurations, managing resources, and supporting system commissioning.

#### Scope

The scope covers the following :

- Pre-installation Planning and Site Readiness
- Team Coordination and Resource Allocation
- Installation Oversight and Configuration
- System Integration and Testing

#### Elements and Performance Criteria

##### *Pre-installation planning and site readiness*

To be competent, the user/individual on the job must be able to:

- PC1.** verify site preparedness by checking for adequate space, ventilation, environmental factors, power supply, and network connectivity to support IoT device deployment.
- PC2.** inspect physical and network infrastructure to ensure compatibility with IoT systems and resolve discrepancies before installation.
- PC3.** supervise the delivery, unpacking, and safe storage of IoT components and peripherals to prevent damage or loss.

##### *Team coordination and resource allocation*

To be competent, the user/individual on the job must be able to:

- PC4.** allocate tasks to technicians and guide them through wiring, mounting, and assembling activities based on installation blueprints or system diagrams.
- PC5.** brief the team on safety procedures, tool usage, and installation protocols, ensuring compliance with electrical and electronic standards.
- PC6.** maintain a checklist of required materials, tools, and documentation to avoid delays in installation.

##### *Installation oversight and configuration*

To be competent, the user/individual on the job must be able to:

- PC7.** oversee the installation of sensors, controllers, gateways, and end-user devices as per manufacturer guidelines and site layout.
- PC8.** validate electrical connections, device positioning, and sensor calibration before initiating the system configuration.
- PC9.** guide the team to perform firmware updates and basic device configuration using the designated user interface or setup software.

##### *System integration and testing*

To be competent, the user/individual on the job must be able to:

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- PC10.** ensure seamless integration of installed devices with backend systems, cloud platforms, or edge devices, following architecture design.
- PC11.** supervise system testing, including connectivity checks, data flow validation, device discovery, and network performance.
- PC12.** identify and resolve minor configuration issues or escalate critical problems to the technical support or R&D team.

### *Documentation and handover*

To be competent, the user/individual on the job must be able to:

- PC13.** maintain installation records including device inventory, configuration files, network settings, and test reports.
- PC14.** document deviations from the original plan and suggest corrective actions or improvements.
- PC15.** hand over the installed system to the operations or maintenance team with complete documentation and walkthrough.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company policies related to installation, safety, and reporting.
- KU2.** standard operating procedures for equipment handling and site coordination.
- KU3.** inter-department collaboration between IT, electrical, civil, and safety teams.
- KU4.** quality control processes for IoT installation projects.
- KU5.** escalation matrix and decision-making authority.
- KU6.** basic architecture of IoT systems, including device, network, and cloud layers.
- KU7.** types of IoT devices (e.g., sensors, actuators, controllers, gateways) and their functions.
- KU8.** electrical wiring and network cabling practices relevant to IoT deployments.
- KU9.** tools and software used for device setup, configuration, and firmware update.
- KU10.** integration methods including MQTT, REST API, HTTP, and relevant protocols.
- KU11.** testing tools and procedures for validating device performance and network connectivity.
- KU12.** documentation techniques and best practices for project tracking and handover.
- KU13.** basic cybersecurity precautions during IoT installation (e.g., secure credentials, firmware integrity).

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret technical manuals, wiring diagrams, and installation guidelines.
- GS2.** communicate effectively with installation teams, vendors, and supervisors.
- GS3.** stay updated on IoT hardware, software protocols, and installation standards.
- GS4.** manage time and resources to ensure timely and accurate installation.
- GS5.** identify and resolve issues related to device compatibility, connectivity, or setup.

## Qualification Pack

- GS6.** apply critical thinking to ensure installation quality, system integrity, and compliance with safety norms.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Pre-installation planning and site readiness</i>                                                                                                                               | <b>6</b>     | <b>12</b>       | -             | <b>2</b>   |
| <b>PC1.</b> verify site preparedness by checking for adequate space, ventilation, environmental factors, power supply, and network connectivity to support IoT device deployment. | 2            | 4               | -             | 1          |
| <b>PC2.</b> inspect physical and network infrastructure to ensure compatibility with IoT systems and resolve discrepancies before installation.                                   | 2            | 4               | -             | -          |
| <b>PC3.</b> supervise the delivery, unpacking, and safe storage of IoT components and peripherals to prevent damage or loss.                                                      | 2            | 4               | -             | 1          |
| <i>Team coordination and resource allocation</i>                                                                                                                                  | <b>6</b>     | <b>12</b>       | -             | <b>2</b>   |
| <b>PC4.</b> allocate tasks to technicians and guide them through wiring, mounting, and assembling activities based on installation blueprints or system diagrams.                 | 2            | 4               | -             | -          |
| <b>PC5.</b> brief the team on safety procedures, tool usage, and installation protocols, ensuring compliance with electrical and electronic standards.                            | 2            | 4               | -             | 1          |
| <b>PC6.</b> maintain a checklist of required materials, tools, and documentation to avoid delays in installation.                                                                 | 2            | 4               | -             | 1          |
| <i>Installation oversight and configuration</i>                                                                                                                                   | <b>6</b>     | <b>12</b>       | -             | <b>2</b>   |
| <b>PC7.</b> oversee the installation of sensors, controllers, gateways, and end-user devices as per manufacturer guidelines and site layout.                                      | 2            | 4               | -             | 1          |
| <b>PC8.</b> validate electrical connections, device positioning, and sensor calibration before initiating the system configuration.                                               | 2            | 4               | -             | 1          |
| <b>PC9.</b> guide the team to perform firmware updates and basic device configuration using the designated user interface or setup software.                                      | 2            | 4               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>System integration and testing</i>                                                                                                                | <b>6</b>     | <b>12</b>       | -             | <b>3</b>   |
| <b>PC10.</b> ensure seamless integration of installed devices with backend systems, cloud platforms, or edge devices, following architecture design. | 2            | 4               | -             | 1          |
| <b>PC11.</b> supervise system testing, including connectivity checks, data flow validation, device discovery, and network performance.               | 2            | 4               | -             | 1          |
| <b>PC12.</b> identify and resolve minor configuration issues or escalate critical problems to the technical support or R&D team.                     | 2            | 4               | -             | 1          |
| <i>Documentation and handover</i>                                                                                                                    | <b>6</b>     | <b>12</b>       | -             | <b>1</b>   |
| <b>PC13.</b> maintain installation records including device inventory, configuration files, network settings, and test reports.                      | 2            | 4               | -             | -          |
| <b>PC14.</b> document deviations from the original plan and suggest corrective actions or improvements.                                              | 2            | 4               | -             | -          |
| <b>PC15.</b> hand over the installed system to the operations or maintenance team with complete documentation and walkthrough.                       | 2            | 4               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                     | <b>30</b>    | <b>60</b>       | -             | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                  |
|----------------------------|--------------------------------------------------|
| <b>NOS Code</b>            | TEL/N6261                                        |
| <b>NOS Name</b>            | Supervise in Installation IoT Devices and System |
| <b>Sector</b>              | Telecom                                          |
| <b>Sub-Sector</b>          | Network Managed Services                         |
| <b>Occupation</b>          | Network Operation and Maintenance                |
| <b>NSQF Level</b>          | 5                                                |
| <b>Credits</b>             | 3                                                |
| <b>Version</b>             | 3.0                                              |
| <b>Last Reviewed Date</b>  | 19/08/2025                                       |
| <b>Next Review Date</b>    | 30/06/2028                                       |
| <b>NSQC Clearance Date</b> | 19/08/2025                                       |

## Qualification Pack

# TEL/N6262: Administer Acceptance Testing and Site Optimization Activities

## Description

The OS oversees acceptance testing and site optimization for IoT systems, ensuring they meet client and organizational standards. They prepare test protocols, execute validation, identify non-conformities, and recommend optimizations.

## Scope

The scope covers the following :

- Planning and preparation for acceptance testing
- Executing acceptance testing
- Defect identification and reporting
- Site optimization and performance tuning
- Documentation, sign-off, and handover

## Elements and Performance Criteria

### *Planning and preparation for acceptance testing*

To be competent, the user/individual on the job must be able to:

- PC1.** review project specifications, acceptance criteria, and system documentation to understand expected outcomes for site testing.
- PC2.** coordinate with installation, integration, and quality teams to schedule acceptance testing without disrupting operations.
- PC3.** prepare testing tools, checklists, and verification protocols required to conduct system validation efficiently and safely.

### *Executing acceptance testing*

To be competent, the user/individual on the job must be able to:

- PC4.** conduct functional, performance, and interoperability tests to verify IoT device behaviour and system response under real-world conditions.
- PC5.** validate connectivity, data integrity, latency, power efficiency, and security configurations as per project requirements.
- PC6.** record all test results, observations, and deviations systematically for review and documentation.

### *Defect identification and reporting*

To be competent, the user/individual on the job must be able to:

- PC7.** identify and document any failures, bugs, or non-conformities found during testing using standardized formats.
- PC8.** escalate critical issues or discrepancies to the engineering or vendor team for resolution.
- PC9.** recommend corrective actions or changes in configuration, firmware, or deployment strategy based on test results.

### *Site optimization and performance tuning*

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To be competent, the user/individual on the job must be able to:

- PC10.** analyze system logs, signal strength, latency data, and throughput to identify areas needing optimization.
- PC11.** fine-tune device settings, network parameters, or sensor placement to improve system efficiency and reliability.
- PC12.** revalidate performance post-optimization to ensure improvements align with predefined thresholds.

### *Documentation, sign-off, and handover*

To be competent, the user/individual on the job must be able to:

- PC13.** compile detailed acceptance test reports, optimization logs, and updated site diagrams for review.
- PC14.** obtain formal sign-off from clients or quality teams after successful acceptance and optimization.
- PC15.** support handover by providing manuals, training, and documentation to operations or maintenance teams.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and quality standards for acceptance testing and deployment sign-off.
- KU2.** types of acceptance tests such as functional, performance, stress, regression, and interoperability testing.
- KU3.** test planning elements including test cases, test scripts, data sets, and environment preparation.
- KU4.** parameters for evaluating IoT systems such as data accuracy, latency, uptime, signal strength, and power efficiency.
- KU5.** techniques for identifying, logging, and categorizing bugs or non-conformities during site testing.
- KU6.** escalation procedures and stakeholder roles for issue resolution and testing sign-offs.
- KU7.** tools and utilities used for testing, monitoring, and logging performance of IoT devices and networks (e.g., Wireshark, Postman, MQTT.fx).
- KU8.** optimization techniques such as device reconfiguration, antenna alignment, firmware tuning, and sensor placement.
- KU9.** compliance requirements and performance benchmarks set by clients or regulatory bodies.
- KU10.** documentation practices for creating test reports, defect logs, optimization records, and handover manuals.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret test cases, system documentation, technical specifications, and client acceptance criteria.

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- GS2.** write structured test reports, defect logs, and technical summaries clearly and concisely.
- GS3.** communicate effectively with cross-functional teams, clients, and supervisors to coordinate test plans and report findings.
- GS4.** use logical reasoning and problem-solving to diagnose performance issues and recommend optimizations.
- GS5.** plan and manage time efficiently to conduct testing and optimization within project deadlines.
- GS6.** demonstrate precision in test measurements, result logging, and device configuration.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Planning and preparation for acceptance testing</i>                                                                                                  | <b>6</b>     | <b>12</b>       | -             | <b>2</b>   |
| <b>PC1.</b> review project specifications, acceptance criteria, and system documentation to understand expected outcomes for site testing.              | 2            | 4               | -             | 1          |
| <b>PC2.</b> coordinate with installation, integration, and quality teams to schedule acceptance testing without disrupting operations.                  | 2            | 4               | -             | -          |
| <b>PC3.</b> prepare testing tools, checklists, and verification protocols required to conduct system validation efficiently and safely.                 | 2            | 4               | -             | 1          |
| <i>Executing acceptance testing</i>                                                                                                                     | <b>6</b>     | <b>12</b>       | -             | <b>2</b>   |
| <b>PC4.</b> conduct functional, performance, and interoperability tests to verify IoT device behaviour and system response under real-world conditions. | 2            | 4               | -             | -          |
| <b>PC5.</b> validate connectivity, data integrity, latency, power efficiency, and security configurations as per project requirements.                  | 2            | 4               | -             | 1          |
| <b>PC6.</b> record all test results, observations, and deviations systematically for review and documentation.                                          | 2            | 4               | -             | 1          |
| <i>Defect identification and reporting</i>                                                                                                              | <b>6</b>     | <b>12</b>       | -             | <b>2</b>   |
| <b>PC7.</b> identify and document any failures, bugs, or non-conformities found during testing using standardized formats.                              | 2            | 4               | -             | 1          |
| <b>PC8.</b> escalate critical issues or discrepancies to the engineering or vendor team for resolution.                                                 | 2            | 4               | -             | 1          |
| <b>PC9.</b> recommend corrective actions or changes in configuration, firmware, or deployment strategy based on test results.                           | 2            | 4               | -             | -          |
| <i>Site optimization and performance tuning</i>                                                                                                         | <b>6</b>     | <b>12</b>       | -             | <b>3</b>   |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> analyze system logs, signal strength, latency data, and throughput to identify areas needing optimization.       | 2            | 4               | -             | 1          |
| <b>PC11.</b> fine-tune device settings, network parameters, or sensor placement to improve system efficiency and reliability. | 2            | 4               | -             | 1          |
| <b>PC12.</b> revalidate performance post-optimization to ensure improvements align with predefined thresholds.                | 2            | 4               | -             | 1          |
| <i>Documentation, sign-off, and handover</i>                                                                                  | <b>6</b>     | <b>12</b>       | -             | <b>1</b>   |
| <b>PC13.</b> compile detailed acceptance test reports, optimization logs, and updated site diagrams for review.               | 2            | 4               | -             | -          |
| <b>PC14.</b> obtain formal sign-off from clients or quality teams after successful acceptance and optimization.               | 2            | 4               | -             | -          |
| <b>PC15.</b> support handover by providing manuals, training, and documentation to operations or maintenance teams.           | 2            | 4               | -             | 1          |
| <b>NOS Total</b>                                                                                                              | <b>30</b>    | <b>60</b>       | -             | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                |
|----------------------------|----------------------------------------------------------------|
| <b>NOS Code</b>            | TEL/N6262                                                      |
| <b>NOS Name</b>            | Administer Acceptance Testing and Site Optimization Activities |
| <b>Sector</b>              | Telecom                                                        |
| <b>Sub-Sector</b>          | Network Managed Services                                       |
| <b>Occupation</b>          | Network Operation and Maintenance                              |
| <b>NSQF Level</b>          | 3                                                              |
| <b>Credits</b>             | 3                                                              |
| <b>Version</b>             | 3.0                                                            |
| <b>Last Reviewed Date</b>  | 19/08/2025                                                     |
| <b>Next Review Date</b>    | 30/06/2028                                                     |
| <b>NSQC Clearance Date</b> | 19/08/2025                                                     |

## Qualification Pack

### TEL/N6280: Maintain Secure IoT Network Infrastructure

#### Description

This OS focuses on secure, scalable, and efficient IoT network infrastructure, evaluating application requirements, selecting appropriate technologies, ensuring data protection, mitigating risks, and documenting architecture compliance.

#### Scope

The scope covers the following :

- Requirement analysis and planning
- Security architecture and configuration
- Risk mitigation and compliance
- Testing, optimization, and documentation

#### Elements and Performance Criteria

##### *Requirement analysis and planning*

To be competent, the user/individual on the job must be able to:

- PC1.** assess device and application-specific networking needs such as range, latency, bandwidth, and number of nodes.
- PC2.** select appropriate communication technologies (e.g., Wi-Fi, LoRaWAN, Zigbee, NB-IoT, Ethernet) based on deployment conditions and data requirements.
- PC3.** define IP addressing schemes, subnetting plans, and device naming conventions to ensure network scalability and manageability.
- PC4.** assist in designing logical and physical network topology (e.g., star, mesh, hybrid) to suit system architecture and operational efficiency.

##### *Security architecture and configuration*

To be competent, the user/individual on the job must be able to:

- PC5.** implement encryption protocols (e.g., TLS, IPSec, WPA3) and secure tunneling to protect data during transmission.
- PC6.** configure firewalls, VLANs, and access control policies to restrict unauthorized access and isolate IoT segments from general networks.
- PC7.** deploy authentication mechanisms such as certificates, tokens, or key-based access for devices and gateways.
- PC8.** integrate intrusion detection/prevention systems (IDS/IPS) and secure update mechanisms to maintain system integrity.

##### *Risk mitigation and compliance*

To be competent, the user/individual on the job must be able to:

- PC9.** identify potential vulnerabilities such as open ports, weak credentials, or outdated firmware and apply corrective configurations.
- PC10.** ensure adherence to regulatory requirements and data protection standards like ISO 27001, NIST IoT, and GDPR.

## Qualification Pack

**PC11.** develop contingency and recovery plans for network failures, data breaches, or cyberattacks.

*Testing, optimization, and documentation*

To be competent, the user/individual on the job must be able to:

**PC12.** test network for performance indicators such as throughput, signal strength, latency, and packet loss under various load conditions.

**PC13.** simulate cyber-threat scenarios (e.g., spoofing, sniffing, DDoS) to validate security mechanisms.

**PC14.** document the entire network architecture, device configurations, security protocols, and compliance mappings for audit readiness.

**PC15.** prepare user manuals and technical documentation for use by support and maintenance teams.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** organizational IT and network security policies, standards, and escalation protocols.

**KU2.** IoT-specific communication technologies and protocols (Wi-Fi, Zigbee, LoRaWAN, NB-IoT, BLE, MQTT, CoAP, HTTP).

**KU3.** IP addressing, subnetting, VLANs, routing techniques, NAT, and QoS in IoT environments.

**KU4.** encryption techniques and secure communication protocols (TLS, SSL, WPA3, IPSec).

**KU5.** firewall configurations, access control lists (ACLs), and zero-trust network segmentation.

**KU6.** vulnerabilities common to IoT systems and strategies for mitigation (e.g., firmware integrity, secure boot, OTA updates).

**KU7.** compliance and regulatory standards like ISO 27001, GDPR, NIST for IoT networks.

**KU8.** tools for network simulation, penetration testing, performance monitoring, and security validation.

**KU9.** documentation practices, audit preparation, and handover protocols for network systems.

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** read and interpret network architecture diagrams, security policies, and protocol specifications.

**GS2.** communicate network design requirements and security considerations with team members and stakeholders.

**GS3.** stay updated on the latest IoT security threats, encryption standards, and network protocols.

**GS4.** plan and manage tasks to ensure timely deployment of secure and scalable network infrastructure.

**GS5.** identify and mitigate potential vulnerabilities in network design and data transmission.

**GS6.** apply critical thinking to balance performance, scalability, and security in network infrastructure design.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Requirement analysis and planning</i>                                                                                                                         | <b>8</b>     | <b>16</b>       | -             | <b>2</b>   |
| <b>PC1.</b> assess device and application-specific networking needs such as range, latency, bandwidth, and number of nodes.                                      | 2            | 4               | -             | -          |
| <b>PC2.</b> select appropriate communication technologies (e.g., Wi-Fi, LoRaWAN, Zigbee, NB-IoT, Ethernet) based on deployment conditions and data requirements. | 2            | 4               | -             | -          |
| <b>PC3.</b> define IP addressing schemes, subnetting plans, and device naming conventions to ensure network scalability and manageability.                       | 2            | 4               | -             | 1          |
| <b>PC4.</b> assist in designing logical and physical network topology (e.g., star, mesh, hybrid) to suit system architecture and operational efficiency.         | 2            | 4               | -             | 1          |
| <i>Security architecture and configuration</i>                                                                                                                   | <b>8</b>     | <b>16</b>       | -             | <b>3</b>   |
| <b>PC5.</b> implement encryption protocols (e.g., TLS, IPSec, WPA3) and secure tunneling to protect data during transmission.                                    | 2            | 4               | -             | 1          |
| <b>PC6.</b> configure firewalls, VLANs, and access control policies to restrict unauthorized access and isolate IoT segments from general networks.              | 2            | 4               | -             | -          |
| <b>PC7.</b> deploy authentication mechanisms such as certificates, tokens, or key-based access for devices and gateways.                                         | 2            | 4               | -             | 1          |
| <b>PC8.</b> integrate intrusion detection/prevention systems (IDS/IPS) and secure update mechanisms to maintain system integrity.                                | 2            | 4               | -             | 1          |
| <i>Risk mitigation and compliance</i>                                                                                                                            | <b>6</b>     | <b>12</b>       | -             | <b>1</b>   |
| <b>PC9.</b> identify potential vulnerabilities such as open ports, weak credentials, or outdated firmware and apply corrective configurations.                   | 2            | 4               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> ensure adherence to regulatory requirements and data protection standards like ISO 27001, NIST IoT, and GDPR.                        | 2            | 4               | -             | -          |
| <b>PC11.</b> develop contingency and recovery plans for network failures, data breaches, or cyberattacks.                                         | 2            | 4               | -             | 1          |
| <i>Testing, optimization, and documentation</i>                                                                                                   | <b>8</b>     | <b>16</b>       | -             | <b>4</b>   |
| <b>PC12.</b> test network for performance indicators such as throughput, signal strength, latency, and packet loss under various load conditions. | 2            | 4               | -             | 1          |
| <b>PC13.</b> simulate cyber-threat scenarios (e.g., spoofing, sniffing, DDoS) to validate security mechanisms.                                    | 2            | 4               | -             | 1          |
| <b>PC14.</b> document the entire network architecture, device configurations, security protocols, and compliance mappings for audit readiness.    | 2            | 4               | -             | 1          |
| <b>PC15.</b> prepare user manuals and technical documentation for use by support and maintenance teams.                                           | 2            | 4               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                  | <b>30</b>    | <b>60</b>       | -             | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                            |
|----------------------------|--------------------------------------------|
| <b>NOS Code</b>            | TEL/N6280                                  |
| <b>NOS Name</b>            | Maintain Secure IoT Network Infrastructure |
| <b>Sector</b>              | Telecom                                    |
| <b>Sub-Sector</b>          |                                            |
| <b>Occupation</b>          | Network Operation and Maintenance          |
| <b>NSQF Level</b>          | 5                                          |
| <b>Credits</b>             | 4                                          |
| <b>Version</b>             | 1.0                                        |
| <b>Last Reviewed Date</b>  | 19/08/2025                                 |
| <b>Next Review Date</b>    | 30/06/2028                                 |
| <b>NSQC Clearance Date</b> | 19/08/2025                                 |

## Qualification Pack

### TEL/N9103: Implement Effective Interaction at workplace

#### Description

This OS unit is about communicating with superiors, colleagues, customers, and other stakeholders in a respectful, inclusive, and productive manner.

#### Scope

The scope covers the following :

- Interact effectively with supervisors and stakeholders
- Collaborate efficiently with colleagues and customers
- Practice inclusive communication at workplace

#### Elements and Performance Criteria

##### *Interact effectively with supervisors and stakeholders*

To be competent, the user/individual on the job must be able to:

- PC1.** interpret and clarify work requirements and expectations from supervisors or stakeholders
- PC2.** report any delays, risks, or unforeseen disruptions to the appropriate authority promptly
- PC3.** deliver work outputs as per quality, productivity, and timeline standards defined by the organisation.

##### *Collaborate efficiently with colleagues and customers*

To be competent, the user/individual on the job must be able to:

- PC4.** share work requirements and expectations clearly with team members
- PC5.** communicate using appropriate channels (e.g., face-to-face, phone, e-mail, video conferencing, chat) based on context
- PC6.** coordinate effectively with others to integrate efforts and achieve collective goals
- PC7.** respect colleagues' and customers' personal and professional boundaries
- PC8.** collaboratively solve work-related issues or bottlenecks with mutual consent
- PC9.** resolve interpersonal or operational conflicts within the team professionally and constructively
- PC10.** motivate team members to align with organisational objectives over individual interests
- PC11.** encourage two-way feedback and constructive dialogue within the team.

##### *Practice inclusive communication at workplace*

To be competent, the user/individual on the job must be able to:

- PC12.** ensure own and team's conduct reflects respect for gender, disability, and diversity in all interactions
- PC13.** communicate with sensitivity toward the needs of persons with disabilities (PwD) and across gender identities
- PC14.** demonstrate awareness of different types of disabilities and their implications in workplace communication
- PC15.** assist team members with disabilities in overcoming barriers to work

## Qualification Pack

**PC16.** use inclusive and respectful language in all verbal and written communication

**PC17.** treat all individuals with dignity and fairness, avoiding bias or discrimination

**PC18.** respect others' personal space, cultural preferences, and accessibility needs.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** importance of effective communication and team collaboration in achieving workplace goals

**KU2.** modes and tools of communication for various scenarios (in-person, virtual, asynchronous)

**KU3.** organisational processes for task assignment, reporting, and escalation

**KU4.** impact of one's work on the team's workflow and deliverables

**KU5.** strategies to handle team dynamics and conflict resolution

**KU6.** principles of inclusive behaviour and workplace diversity

**KU7.** basic concepts of gender sensitivity, unconscious bias, and respectful workplace behaviour

**KU8.** categories of disabilities and associated workplace considerations

**KU9.** rights, entitlements, and accommodations applicable to PwDs under national and organisational policies

**KU10.** health, safety, and digital accessibility requirements for inclusive workplaces

**KU11.** government and private schemes supporting employment and inclusion of PwDs

**KU12.** practices to encourage inclusive recruitment and professional growth

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** communicate clearly and respectfully using verbal, non-verbal, and written modes.

**GS2.** maintain professionalism in customer and colleague interactions

**GS3.** identify, address, or escalate interpersonal challenges constructively

**GS4.** impact of reflect on one's biases, assumptions, and inclusive practices

**GS5.** empathise with and advocate for individuals from underrepresented groups

**GS6.** promote collaboration and resolve conflicts through dialogue and consensus

**GS7.** engage in continuous improvement of interpersonal and team dynamics

**GS8.** use digital communication tools responsibly and inclusively in a hybrid work setting.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Interact effectively with supervisors and stakeholders</i>                                                                     | <b>5</b>     | <b>10</b>       | -             | <b>1</b>   |
| <b>PC1.</b> interpret and clarify work requirements and expectations from supervisors or stakeholders                             | 2            | 3               | -             | -          |
| <b>PC2.</b> report any delays, risks, or unforeseen disruptions to the appropriate authority promptly                             | 2            | 4               | -             | 1          |
| <b>PC3.</b> deliver work outputs as per quality, productivity, and timeline standards defined by the organisation.                | 1            | 3               | -             | -          |
| <i>Collaborate efficiently with colleagues and customers</i>                                                                      | <b>14</b>    | <b>26</b>       | -             | <b>6</b>   |
| <b>PC4.</b> share work requirements and expectations clearly with team members                                                    | 2            | 3               | -             | 1          |
| <b>PC5.</b> communicate using appropriate channels (e.g., face-to-face, phone, e-mail, video conferencing, chat) based on context | 2            | 4               | -             | 1          |
| <b>PC6.</b> coordinate effectively with others to integrate efforts and achieve collective goals                                  | 2            | 3               | -             | -          |
| <b>PC7.</b> respect colleagues' and customers' personal and professional boundaries                                               | 2            | 3               | -             | 1          |
| <b>PC8.</b> collaboratively solve work-related issues or bottlenecks with mutual consent                                          | 2            | 3               | -             | 1          |
| <b>PC9.</b> resolve interpersonal or operational conflicts within the team professionally and constructively                      | 1            | 3               | -             | 1          |
| <b>PC10.</b> motivate team members to align with organisational objectives over individual interests                              | 2            | 4               | -             | -          |
| <b>PC11.</b> encourage two-way feedback and constructive dialogue within the team.                                                | 1            | 3               | -             | 1          |
| <i>Practice inclusive communication at workplace</i>                                                                              | <b>11</b>    | <b>24</b>       | -             | <b>3</b>   |
| <b>PC12.</b> ensure own and team's conduct reflects respect for gender, disability, and diversity in all interactions             | 2            | 3               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> communicate with sensitivity toward the needs of persons with disabilities (PwD) and across gender identities | 2            | 4               | -             | 1          |
| <b>PC14.</b> demonstrate awareness of different types of disabilities and their implications in workplace communication    | 1            | 4               | -             | -          |
| <b>PC15.</b> assist team members with disabilities in overcoming barriers to work                                          | 1            | 4               | -             | -          |
| <b>PC16.</b> use inclusive and respectful language in all verbal and written communication                                 | 2            | 3               | -             | -          |
| <b>PC17.</b> treat all individuals with dignity and fairness, avoiding bias or discrimination                              | 1            | 3               | -             | 1          |
| <b>PC18.</b> respect others' personal space, cultural preferences, and accessibility needs.                                | 2            | 3               | -             | 1          |
| <b>NOS Total</b>                                                                                                           | <b>30</b>    | <b>60</b>       | <b>-</b>      | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                              |
|----------------------------|----------------------------------------------|
| <b>NOS Code</b>            | TEL/N9103                                    |
| <b>NOS Name</b>            | Implement Effective Interaction at workplace |
| <b>Sector</b>              | Telecom                                      |
| <b>Sub-Sector</b>          | Generic                                      |
| <b>Occupation</b>          | Generic                                      |
| <b>NSQF Level</b>          | 5                                            |
| <b>Credits</b>             | 1                                            |
| <b>Version</b>             | 3.0                                          |
| <b>Last Reviewed Date</b>  | 19/08/2025                                   |
| <b>Next Review Date</b>    | 30/06/2028                                   |
| <b>NSQC Clearance Date</b> | 19/08/2025                                   |

## Qualification Pack

### TEL/N9104: Manage Work, Resources and Safety at workplace

#### Description

This OS unit is about applying sustainable practices, maintaining workplace safety, and ensuring optimal utilisation of resources to meet quality standards and deadlines in a dynamic work environment.

#### Scope

The scope covers the following :

- Manage learning and self-direction
- Foster critical thinking and problem-solving
- Perform work as per quality and performance standards
- Maintain a safe, secure, and healthy workplace
- Implement sustainable and efficient resource usage practices

#### Elements and Performance Criteria

##### *Manage learning and self-direction*

To be competent, the user/individual on the job must be able to:

- PC1.** identify individual and team training needs relevant to current and emerging job roles
- PC2.** participate in learning activities and skill development programs
- PC3.** encourage team participation in professional development, cross-functional knowledge sharing, and upskilling initiatives.

##### *Foster critical thinking and problem-solving*

To be competent, the user/individual on the job must be able to:

- PC4.** guide team members to analyse problems logically and identify root causes
- PC5.** suggest appropriate and timely solutions to operational challenges
- PC6.** enable the team to evaluate outcomes of decisions and refine processes accordingly.

##### *Perform work as per quality and performance standards*

To be competent, the user/individual on the job must be able to:

- PC7.** organise individual and team work based on task priorities and deadlines
- PC8.** monitor performance to ensure quality, accuracy, and timely completion of tasks
- PC9.** allocate responsibilities and establish work schedules to optimise team productivity
- PC10.** implement systems to track adherence to standard operating procedures (SOPs)

##### *Maintain a safe, secure, and healthy workplace*

To be competent, the user/individual on the job must be able to:

- PC11.** communicate safety policies and procedures clearly to all team members
- PC12.** address risks such as accidents, electrical hazards, cyber risks, or health incidents as per emergency protocols
- PC13.** report hazards, breaches, or incidents promptly to the designated authority
- PC14.** promote practices that support mental well-being and respectful workplace behaviour.

##### *Implement sustainable and efficient resource usage practices*

## Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC15.** implement ways to optimise the usage of material, energy and water, in various tasks/activities/processes
- PC16.** encourage regular upkeep of tools, machinery, and digital assets to extend usability
- PC17.** ensure team compliance with environmentally responsible work practices
- PC18.** promote the use of energy-efficient systems and paperless documentation where feasible.
- PC19.** report and resolve malfunctions or wastage through preventive or corrective actions

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles of continuous learning and knowledge management
- KU2.** strategies pertinent to the field that can be used to pursue an advancement of skills
- KU3.** organisational policies on performance standards and task completion
- KU4.** methods for identifying training and development needs
- KU5.** problem-solving frameworks and techniques
- KU6.** quality assurance systems and key performance indicators (KPIs)
- KU7.** health, safety, and cybersecurity practices applicable to the work environment
- KU8.** types of workplace hazards (physical, electrical, digital, etc.) and emergency response protocols
- KU9.** ways to conserve energy, water, and materials during operations
- KU10.** routine maintenance procedures for tools, equipment, or digital infrastructure
- KU11.** environmental, social, and governance (ESG) principles relevant to workplace operations.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** apply strategies for continuous improvement in personal and team performance.
- GS2.** think critically and adapt to changing work scenarios
- GS3.** interpret feedback from superiors in a constructive way
- GS4.** communicate effectively with team members, supervisors, and stakeholders
- GS5.** organise and manage work plans, schedules, and deliverables
- GS6.** interpret and act on feedback for professional growth
- GS7.** escalate safety, operational, or ethical issues to the appropriate authority
- GS8.** promote inclusive and respectful team dynamics
- GS9.** take ownership of work outcomes and support team accountability

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Manage learning and self-direction</i>                                                                                             | <b>7</b>     | <b>9</b>        | -             | <b>2</b>   |
| <b>PC1.</b> identify individual and team training needs relevant to current and emerging job roles                                    | 3            | 3               | -             | 1          |
| <b>PC2.</b> participate in learning activities and skill development programs                                                         | 2            | 3               | -             | -          |
| <b>PC3.</b> encourage team participation in professional development, cross-functional knowledge sharing, and upskilling initiatives. | 2            | 3               | -             | 1          |
| <i>Foster critical thinking and problem-solving</i>                                                                                   | <b>6</b>     | <b>6</b>        | -             | <b>1</b>   |
| <b>PC4.</b> guide team members to analyse problems logically and identify root causes                                                 | 2            | 2               | -             | -          |
| <b>PC5.</b> suggest appropriate and timely solutions to operational challenges                                                        | 2            | 2               | -             | 1          |
| <b>PC6.</b> enable the team to evaluate outcomes of decisions and refine processes accordingly.                                       | 2            | 2               | -             | -          |
| <i>Perform work as per quality and performance standards</i>                                                                          | <b>9</b>     | <b>10</b>       | -             | <b>2</b>   |
| <b>PC7.</b> organise individual and team work based on task priorities and deadlines                                                  | 3            | 3               | -             | -          |
| <b>PC8.</b> monitor performance to ensure quality, accuracy, and timely completion of tasks                                           | 2            | 3               | -             | -          |
| <b>PC9.</b> allocate responsibilities and establish work schedules to optimise team productivity                                      | 2            | 2               | -             | 1          |
| <b>PC10.</b> implement systems to track adherence to standard operating procedures (SOPs)                                             | 2            | 2               | -             | 1          |
| <i>Maintain a safe, secure, and healthy workplace</i>                                                                                 | <b>8</b>     | <b>11</b>       | -             | <b>3</b>   |
| <b>PC11.</b> communicate safety policies and procedures clearly to all team members                                                   | 2            | 3               | -             | 1          |
| <b>PC12.</b> address risks such as accidents, electrical hazards, cyber risks, or health incidents as per emergency protocols         | 2            | 3               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> report hazards, breaches, or incidents promptly to the designated authority                               | 2            | 3               | -             | 1          |
| <b>PC14.</b> promote practices that support mental well-being and respectful workplace behaviour.                      | 2            | 2               | -             | 1          |
| <i>Implement sustainable and efficient resource usage practices</i>                                                    | <b>10</b>    | <b>14</b>       | -             | <b>2</b>   |
| <b>PC15.</b> implement ways to optimise the usage of material, energy and water, in various tasks/activities/processes | 2            | 3               | -             | -          |
| <b>PC16.</b> encourage regular upkeep of tools, machinery, and digital assets to extend usability                      | 2            | 2               | -             | -          |
| <b>PC17.</b> ensure team compliance with environmentally responsible work practices                                    | 2            | 3               | -             | 1          |
| <b>PC18.</b> promote the use of energy-efficient systems and paperless documentation where feasible.                   | 2            | 3               | -             | 1          |
| <b>PC19.</b> report and resolve malfunctions or wastage through preventive or corrective actions                       | 2            | 3               | -             | -          |
| <b>NOS Total</b>                                                                                                       | <b>40</b>    | <b>50</b>       | -             | <b>10</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                |
|----------------------------|------------------------------------------------|
| <b>NOS Code</b>            | TEL/N9104                                      |
| <b>NOS Name</b>            | Manage Work, Resources and Safety at workplace |
| <b>Sector</b>              | Telecom                                        |
| <b>Sub-Sector</b>          | Generic                                        |
| <b>Occupation</b>          | Generic                                        |
| <b>NSQF Level</b>          | 5                                              |
| <b>Credits</b>             | 1                                              |
| <b>Version</b>             | 3.0                                            |
| <b>Last Reviewed Date</b>  | 19/08/2025                                     |
| <b>Next Review Date</b>    | 30/06/2028                                     |
| <b>NSQC Clearance Date</b> | 19/08/2025                                     |

## Qualification Pack

### DGT/VSQ/N0102: Employability Skills (60 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

## Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

### *Career Development & Goal Setting*

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

### *Communication Skills*

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

### *Customer Service*

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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**PC28.** follow appropriate hygiene and grooming standards

*Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC29.** create a professional Curriculum vitae (Résumé)

**PC30.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

**PC31.** apply to identified job openings using offline /online methods as per requirement

**PC32.** answer questions politely, with clarity and confidence, during recruitment and selection

**PC33.** identify apprenticeship opportunities and register for it as per guidelines and requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills and different learning and employability related portals

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

**KU6.** importance of career development and setting long- and short-term goals

**KU7.** about effective communication

**KU8.** POSH Act

**KU9.** Gender sensitivity and inclusivity

**KU10.** different types of financial institutes, products, and services

**KU11.** how to compute income and expenditure

**KU12.** importance of maintaining safety and security in offline and online financial transactions

**KU13.** different legal rights and laws

**KU14.** different types of digital devices and the procedure to operate them safely and securely

**KU15.** how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

**KU16.** how to identify business opportunities

**KU17.** types and needs of customers

**KU18.** how to apply for a job and prepare for an interview

**KU19.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** read and write different types of documents/instructions/correspondence

**GS2.** communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                                                                      | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> identify employability skills required for jobs in various industries                                                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC2.</b> identify and explore learning and employability portals                                                                                                                                                                                                                                              | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC3.</b> recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.                                                               | -            | -               | -             | -          |
| <b>PC4.</b> follow environmentally sustainable practices                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                                                                               | <b>2</b>     | <b>4</b>        | -             | -          |
| <b>PC5.</b> recognize the significance of 21st Century Skills for employment                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC6.</b> practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                                                                      | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC7.</b> use basic English for everyday conversation in different contexts, in person and over the telephone                                                                                                                                                                                                  | -            | -               | -             | -          |
| <b>PC8.</b> read and understand routine information, notes, instructions, mails, letters etc. written in English                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC9.</b> write short messages, notes, letters, e-mails etc. in English                                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <i>Career Development &amp; Goal Setting</i>                                                                                                                                                                                                                                                                     | <b>1</b>     | <b>2</b>        | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> understand the difference between job and career                                                         | -            | -               | -             | -          |
| <b>PC11.</b> prepare a career development plan with short- and long-term goals, based on aptitude                     | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC12.</b> follow verbal and non-verbal communication etiquette and active listening techniques in various settings | -            | -               | -             | -          |
| <b>PC13.</b> work collaboratively with others in a team                                                               | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                      | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC14.</b> communicate and behave appropriately with all genders and PwD                                            | -            | -               | -             | -          |
| <b>PC15.</b> escalate any issues related to sexual harassment at workplace according to POSH Act                      | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                   | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC16.</b> select financial institutions, products and services as per requirement                                  | -            | -               | -             | -          |
| <b>PC17.</b> carry out offline and online financial transactions, safely and securely                                 | -            | -               | -             | -          |
| <b>PC18.</b> identify common components of salary and compute income, expenses, taxes, investments etc                | -            | -               | -             | -          |
| <b>PC19.</b> identify relevant rights and laws and use legal aids to fight against legal exploitation                 | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                                       | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC20.</b> operate digital devices and carry out basic internet operations securely and safely                      | -            | -               | -             | -          |
| <b>PC21.</b> use e- mail and social media platforms and virtual collaboration tools to work effectively               | -            | -               | -             | -          |
| <b>PC22.</b> use basic features of word processor, spreadsheets, and presentations                                    | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Entrepreneurship</i>                                                                                                                                                          | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC23.</b> identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research                                       | -            | -               | -             | -          |
| <b>PC24.</b> develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion                                                      | -            | -               | -             | -          |
| <b>PC25.</b> identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity                                           | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                                                                                                          | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC26.</b> identify different types of customers                                                                                                                               | -            | -               | -             | -          |
| <b>PC27.</b> identify and respond to customer requests and needs in a professional manner.                                                                                       | -            | -               | -             | -          |
| <b>PC28.</b> follow appropriate hygiene and grooming standards                                                                                                                   | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                                                                                               | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC29.</b> create a professional Curriculum vitae (Résumé)                                                                                                                     | -            | -               | -             | -          |
| <b>PC30.</b> search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively | -            | -               | -             | -          |
| <b>PC31.</b> apply to identified job openings using offline /online methods as per requirement                                                                                   | -            | -               | -             | -          |
| <b>PC32.</b> answer questions politely, with clarity and confidence, during recruitment and selection                                                                            | -            | -               | -             | -          |
| <b>PC33.</b> identify apprenticeship opportunities and register for it as per guidelines and requirements                                                                        | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                 | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0102                   |
| <b>NOS Name</b>            | Employability Skills (60 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 4                               |
| <b>Credits</b>             | 2                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 12/03/2026                      |
| <b>Next Review Date</b>    | 12/03/2029                      |
| <b>NSQC Clearance Date</b> | 12/03/2026                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

## Qualification Pack

### Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

### Assessment Weightage

Compulsory NOS

| National Occupational Standards                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|--------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| TEL/N6260.Perform Market Analysis on Application of IoT                  | 50           | 40              | -             | 10         | 100         | 15         |
| TEL/N6261.Supervise in Installation IoT Devices and System               | 30           | 60              | -             | 10         | 100         | 15         |
| TEL/N6262.Administer Acceptance Testing and Site Optimization Activities | 30           | 60              | -             | 10         | 100         | 15         |
| TEL/N6280.Maintain Secure IoT Network Infrastructure                     | 30           | 60              | -             | 10         | 100         | 15         |
| TEL/N9103.Implement Effective Interaction at workplace                   | 30           | 60              | -             | 10         | 100         | 15         |
| TEL/N9104.Manage Work, Resources and Safety at workplace                 | 40           | 50              | -             | 10         | 100         | 15         |
| DGT/VSQ/N0102.Employability Skills (60 Hours)                            | 20           | 30              | -             | -          | 50          | 10         |
| <b>Total</b>                                                             | <b>230</b>   | <b>360</b>      | <b>-</b>      | <b>60</b>  | <b>650</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualification Pack                              |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |
| <b>Sector</b>                           | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                                                                                            |
| <b>Sub-sector</b>                       | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                                                                                           |
| <b>Occupation</b>                       | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                                                                                      |
| <b>Job role</b>                         | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                                                                                      |
| <b>Occupational Standards (OS)</b>      | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.                                                                       |
| <b>Performance Criteria (PC)</b>        | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                                                                                          |

## Qualification Pack

|                                       |                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>National Occupational Standard</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                                                                                             |
| <b>Qualification Pack (QP)</b>        | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                                                                                             |
| <b>Unit Code</b>                      | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                                                                                              |
| <b>Unit Title</b>                     | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                                                                                              |
| <b>Description</b>                    | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                                                                                         |
| <b>Scope</b>                          | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                                                                                                  |
| <b>Knowledge and Understanding</b>    | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>         | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>            | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills</b>    | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                      | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                        | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |